

Ex 10.1

Question 1.

For each of the following differential equations, determine its order, degree (if exists)

(i) $dydx + xy = \cot x$

Solution:

Order = 1

Degree = 1

(ii) $(d^3ydx^3)^{23} - 3d^2ydx^2 + 5dydx + 4 = 0$

Solution:

$$(d^3ydx^3)^{23} = 3d^2ydx^2 - 5dydx - 4$$

Raising the power to '3' on both the sides

$$(d^3ydx^3)^2 = (3d^2ydx^2 - 5dydx - 4)^3 \quad \text{Order} = 3, \text{ degree} = 2$$

(iii) $(d^2ydx^2)^2 + (dydx)^2 = x \sin(d^2ydx^2)$

Solution:

Highest derivative is d^2ydx^2

$\therefore$ Order = 2

Degree does not exist as d^2ydx^2 cannot be separated from $\sin(d^2ydx^2)$.

(iv) $\sqrt{dydx} - 4dydx - 7x = 0$

Solution:

$$\sqrt{dydx} = 4dydx + 7x$$

Squaring on both sides

$$dydx = (4dydx + 7x)^2$$

Order = 1, degree = 2.

(v)

$$y(dydx) = x(dydx) + (dydx)^3$$

Solution:

$$y(dydx)^2 + y(dydx)^4 = x$$

Order = 1, degree = 4

(vi)

$$x^2 d^2ydx^2 + [1 + (dydx)^2]^{1/2} = 0$$

Solution:

$$x^2 d^2ydx^2 = -[1 + (dydx)^2]^{1/2}$$

Squaring on both sides

$$x^4 (d^2ydx^2)^2 = (1 + (dydx)^2)$$

Order = 2, Degree = 2.

(vii) $(d^2ydx^2)^3 = \sqrt{1 + (dydx)}$

Solution:

Squaring on both sides

$$(d^2ydx^2)^6 = 1 + (dydx) \quad \text{Order} = 2,$$

Degree = 6 (viii) $d^2ydx^2 = xy + \cos(dydx)$ Solution: Order = 2

Since $dydx$ cannot be separated from $\cos(dydx)$, degree does not exist.

(ix) $d^2ydx^2 + 5dydx + \int ydx = x^3$

Solution:

Differentiating both sides

$$d^3ydx^3 + 5d^2ydx^2 + d(\int ydx) = 3x^2 \quad \text{Order} = 3, \text{ Degree} = 1.$$

(x) $X = e^{xy} (dy/dx)$

Solution:

$\log x = xy (dy/dx)$ Order = 1, degree = Not exist

Ex 10.2

Question 1.

Express each of the following physical statements in the form of differential equation.

(i) Radium decays at a rate proportional to the amount Q present.

Solution:

Radium decays at a rate proportional to the amount Q present.

$dQ/dt \propto Q$ $dQ/dt = kQ$ [$\because k$ decay constant]

(ii) The population P of a city increases at a rate proportional to the product of population and to the difference between 5,00,000 and the population.

Solution:

Rate of change of P with respect to ' t ' is dP/dt

Product of population and the difference between 50,000 and the population is $P(50,000 - P)$

Given $dP/dt \propto P(50000 - P)$

$\therefore dP/dt = kP(50000 - P)$ [$\because k$ is a constant]

(iii) For a certain substance, the rate of change of vapor pressure P with respect to temperature T is proportional to the vapor pressure and inversely proportional to the square of the temperature.

Solution:

Rate of change of P with respect to ' t ' is dP/dt

Given, $dP/dt \propto PT^2$

$\therefore dP/dt = kPT^2$ [$\because k$ is a constant]

(iv) A saving amount pays 8% interest per year, compounded continuously. In addition, the income from another investment is credited to the amount continuously at the rate of ₹ 400 per year.

Solution:

Let ' X ' be the amount invested.

Interest = 8% of X

$= 8/100 X = 225X$

$\therefore$ The required equation is $dx/dt = 2x/25 + 400$.

Question 2 .

Assume that a spherical rain drop evaporates at a rate proportional to its surface area. Form a differential equation involving the rate of change of the radius of the rain drop.

Solution:

Volume of sphere $V = \frac{4}{3}\pi r^3$ $dV/dt = 4\pi r^2 \cdot dr/dt$

Surface area $= 4\pi r^2$

Given, $dV/dt \propto (S.A)$

$dV/dt = -k(S.A)$ [k - constant] $4\pi r^2 \cdot dr/dt = -k(4\pi r^2) dr/dt = -k$

Ex 10.3

Question 1.

Find the differential equation of the family of

(i) all non-vertical lines in a plane

Solution:

Equation of family of all non-vertical lines is

$$y = mx + c \quad (m \neq 0)$$

Differentiate with respect to 'x'

$$\frac{dy}{dx} = m \quad ('m' \text{ and } 'c' \text{ are arbitrary constants})$$

Again, $\frac{d^2y}{dx^2} = 0$.

(ii) all non-horizontal lines in a plane.

Solution:

Equation of family of all non-horizontal lines is

$$y = mx + c \quad \frac{dy}{dx} = m \Rightarrow dx dy = 1 m dx^2 \Rightarrow dy^2 = 0.$$

Question 2.

Form the differential equation of all straight lines touching the circle $x^2 + y^2 = r^2$

Solution:

Equation of circle $x^2 + y^2 = r^2$ of the line $y = mx + c$ is to be a tangent to the circle, then the equation of the tangent is

$$y = mx \pm r\sqrt{1+m^2}$$

Differentiating with respect to 'y'

$$\frac{dy}{dx} = m$$

Substitute in (1)

$$y = x\left(\frac{dy}{dx}\right) \pm r\sqrt{1+\left(\frac{dy}{dx}\right)^2} \quad y - x\left(\frac{dy}{dx}\right) = \pm r\sqrt{1+\left(\frac{dy}{dx}\right)^2}$$

Squaring on both sides

$$[y - x\left(\frac{dy}{dx}\right)]^2 = r^2[1 + \left(\frac{dy}{dx}\right)^2] \quad [x\left(\frac{dy}{dx}\right) - y]^2 = r^2[1 + \left(\frac{dy}{dx}\right)^2]$$

Question 3.

Find the differential equation of the family of circles passing through the origin and having their centres on the X-axis.

Solution:

All circles passing through the origin and having their centre on the X-axis say at $(a, 0)$ will have radius 'a' units.

$\therefore$ Equation of circle is $(x - a)^2 + y^2 = a^2 \dots (1)$ [$\because$ 'a' arbitrary constant] Differentiate with respect to 'x'

$$2(x - a) + 2y\frac{dy}{dx} = 0 \quad (\div 2) \quad (x - a) + y\frac{dy}{dx} = 0 \Rightarrow a = x + y\frac{dy}{dx}$$

Substituting in (1)

$$y^2\left(\frac{dy}{dx}\right)^2 + y^2 = (x + y\frac{dy}{dx})^2 \quad y^2\left(\frac{dy}{dx}\right)^2 + y^2 = x^2 + 2xy\frac{dy}{dx} + y^2\left(\frac{dy}{dx}\right)^2 \quad x^2 + 2xy\frac{dy}{dx} - y^2 = 0.$$

Question 4.

Find the differential equation of the family of all the parabolas with latus rectum $4a$ and whose axes are parallel to the X-axis.

Solution:

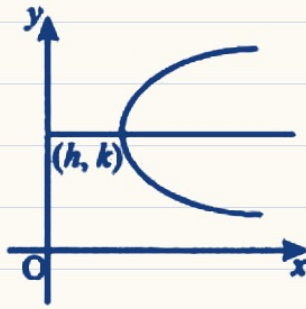
Equation of all parabolas whose axis is parallel to X - axis is $(y - k)^2 = 4a(x - h)$

Where (h, k) is the vertex

$$(y - k)^2 = 4a(x - h)$$

(1) [$\because$ h, k are arbitrary constants]

Differentiate with respect to 'x'



$$2(y-k)dydx=4a \Rightarrow (y-k)dydx=2a \therefore y-k=2a(dy/dx)$$

Now differentiating (2) with respect to 'x'

$$(y-k)d^2ydx^2+(dydx)(dydx)=0 \Rightarrow 2a(dy/dx) \cdot d^2ydx^2+(dydx)^2=0 \quad [\because y-k=2a(dy/dx)]$$

$$2a \cdot (d^2ydx^2)+(dydx)^3=0 \text{ i.e., } 2ay''+y^3=0$$

Question 5.

Find the differential equation of the family of parabolas with vertex at $(0, -1)$ and having axis along the y - axis.

Solution:

Given, vertex $(0, -1)$ and axis along y-axis

Equation of Parabola, $(x+1)^2 = -4ay$

(1) [$\because a$ is the parameter]

Differentiate with respect to 'x'

$$2(x+1) = -4ay' \Rightarrow 2(x+1)y' = -4a$$

Substitute in (1) $\Rightarrow (x+1)^2 = 2(x+1)y'$

$$(x+1) = 2yy'(1+x) \Rightarrow y' - 2y = 0$$

Question 6.

Find the differential equations of the family of all the ellipses having foci on the y-axis and centre at the origin.

Solution:

Equations of the family of all the Ellipses having foci on the y - axis and centre at the origin is

$$x^2/b^2 + y^2/a^2 = 1$$

... (1) [$\because a, b$ are parameters]

Differentiate with respect to 'x'

$$2xb^2 + 2yy'a^2 = 0 (\div 2) \Rightarrow xb^2 + yy'a^2 = 0$$

Differentiate with respect to 'x'

$$1b^2 + 1a^2(yy'' + y'^2) = 0 \Rightarrow 1b^2 = -1a^2(yy'' + y'^2)$$

Substitute in (2) $\Rightarrow -1a^2x(yy'' + y'^2) + yy'a^2 = 0$

$$-xyy'' - xy'^2 + yy' = 0 \Rightarrow xy'^2 + xyy'' - yy' = 0$$

Question 7.

Find the differential equation corresponding to the family of curves represented by the equation

$y = Ae^{8x} + Be^{-8x}$, where A and B are arbitrary constants.

Solution:

$y = Ae^{8x} + Be^{-8x}$ where A and B are arbitrary constants.

Differentiate with respect to 'x'

$$dy/dx = 8Ae^{8x} - 8Be^{-8x}$$

Differentiate again with respect to 'x'

$$d^2y/dx^2 = 64Ae^{8x} + 64Be^{-8x} = 64(Ae^{8x} + Be^{-8x}) \quad d^2y/dx^2 = 64y$$

Question 8.

Find the differential equation of the curve represented by $xy = ae^x + be^{-x} + x^2$.

Solution:

$$xy = ae^x + be^{-x} + x^2 \quad xy - x^2 = ae^x + be^{-x}$$

(1) [$\because a, b$ are arbitrary constants]

Differentiate with respect to 'x'

$$xy' + y - 2x = ae^x - be^{-x}$$

Again, Differentiate with respect to 'x'

$$xy'' + y' + y' - 2 = ae^x + be^{-x} \quad xy'' + 2y' - 2 = ae^x + be^{-x} \quad xy'' + 2y' - 2 = xy - x^2 \quad [\because \text{from (1)}] \quad xy'' + 2y' + x^2 - xy - 2 = 0$$

Ex 10.4

Question 1.

Show that each of the following expressions is a solution of the corresponding given differential equation.

(i) $y = 2x^2$; $xy' = 2y$

Solution:

$$y = 2x^2$$

Differential equation: $xy' = 2y$

Differentiate with respect to 'x'

$$y' = 4x \Rightarrow x = y/4$$

Substitute in (1)

$$y = 2x(x) \quad y = 2x(y/4) \Rightarrow xy = 2$$

On simplifying, $2y = xy'$

$\therefore$ (1) is solution of the given differential equation.

(ii) $y = ae^x + be^{-x}$; $y'' - y = 0$

Solution:

$$y = ae^x + be^{-x} \dots (1) \quad \text{Differential equation: } y'' - y = 0$$

Differentiate with respect to 'x'

$$y' = ae^x - be^{-x}$$

Again differentiate with respect to 'x'

$$y'' = ae^x + be^{-x} \quad y'' = y \Rightarrow y'' - y = 0$$

$\therefore$ (1) is the solution of the given differential equation.

Question 2.

Find value of m so that the function $y = e^{mx}$ is a solution of the given differential equation.

(i) $y + 2y' = 0$

Solution:

Given solution $y = e^{mx}$

Differentiate with respect to 'x'

$$y' = me^{mx} \Rightarrow y' = my$$

$$y' - my = 0$$

Comparing with $y' + 2y = 0 \Rightarrow m = -2$

(ii) $y'' - 5y' + 6y = 0$; $y = e^{mx}$; $y' = me^{mx}$; $y'' = m^2e^{mx}$

Given differential equation is $y'' - 5y' + 6y = 0$

Substitute (1), (2) and (3) in this

$$m^2 e^{mx} - 5m e^{mx} + 6e^{mx} = 0 (\div e^{mx}) \Rightarrow m^2 - 5m + 6 = 0 (m-3)(m-2) = 0 \Rightarrow m = 3, 2$$

Question 3.

The slope of the tangent to the curve at any point is the reciprocal of four times the ordinate at that point. The curve passes through (2,5). Find the equation of the curve.

Solution:

Slope of the tangent is the reciprocal of four times the ordinate

$$\text{i.e., } \frac{dy}{dx} = \frac{1}{4y} \Rightarrow 4y^2 dy = \frac{dx}{4} \Rightarrow 2y^2 = x + c$$

Passes through (2,5)

$$\therefore c = 50 - 2 = 48$$

Question 4.

Show that $y = e^{-x} + mx + n$ is a solution of the differential equation $e^x (d^2y/dx^2) - 1 = 0$.

Solution:

Given solution is $y = e^{-x} + mx + n$

Differentiating (1) with respect to 'x'

$$\frac{dy}{dx} = -e^{-x} + m$$

Again, differentiating with respect to 'x'

$$\frac{d^2y}{dx^2} = e^{-x} = 1 \Rightarrow e^x \left(\frac{d^2y}{dx^2} \right) - 1 = 0$$

Hence proved.

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Question 5.

Show that $y = ax^2 + bx$, $x \neq 0$, is a solution of the differential equation $x^2 y'' + xy' - y = 0$.

Solution:

Given solution : $y = ax^2 + bx$, $x \neq 0$

Here 'a' and 'b' are arbitrary constants

$$y = ax^2 + bx \Rightarrow xy' = 2ax + b$$

$$\text{Differentiate with respect to 'x' } xy' + y \cdot 1 = a(2x) + b \Rightarrow 2ax + b + y = 2ax + b \dots \dots \dots (2)$$

Differentiate again with respect to 'x'.

$$xy'' + y' \cdot 1 + y = 2a \Rightarrow xy'' + 2y' = 2a$$

Substitute (3) in (2)

$$xy'' + y = (xy'' + 2y') - y = 2a - y \Rightarrow xy'' + y = 2a - y \Rightarrow xy'' + 2y = 2a \Rightarrow xy'' + xy' - y = 0$$

Hence proved.

Question 6.

Show that $y = ae^{-3x} + b$, where a and b are arbitrary constants, is a solution of the differential equation

$$d^2y/dx^2 + 3dy/dx = 0.$$

Solution:

$y = ae^{-3x} + b$ (a and b are arbitrary constants)

Differentiate with respect to 'x'

$$\frac{dy}{dx} = -3ae^{-3x}$$

Differentiate again with respect to 'x'

$$\frac{d^2y}{dx^2} = -3(-3ae^{-3x}) = 9ae^{-3x} = -3 \left(\frac{dy}{dx} \right) \Rightarrow \frac{d^2y}{dx^2} + 3 \frac{dy}{dx} = 0$$

Question 7.

$$y^2 = 2a(x + a^2)$$

where a is a

Show that the differential equation representing the family of curves positive parameter, is

Solution:

$$(y^2 - 2xy \frac{dy}{dx})^3 = 8(y \frac{dy}{dx})^5$$

Given equation is $y^2 = 2a(x + a^{2/3})$, 'a' is the parameter.

$$y^2 = 2ax + 2a^{5/3}$$

Differentiate with respect to 'x'

$$2y \frac{dy}{dx} = 2a \Rightarrow y \frac{dy}{dx} = a(1) \Rightarrow y^2 - 2ax = 2a^{5/3}$$

Substitute $a = y \frac{dy}{dx}$

$$y^2 - 2xy \frac{dy}{dx} = 2(y \frac{dy}{dx})^{5/3}$$

Raising the power both sides to '3'

$$(y^2 - 2xy \frac{dy}{dx})^3 = 8(y \frac{dy}{dx})^5$$

Hence proved.

Question 8.

Show that, $y = a \cos bx$ is a solution of the differential equation $\frac{dy}{dx} = -ab \sin bx$

Solution:

$y = a \cos bx \dots (1)$ (a is an arbitrary constant) Differentiating with respect to 'x'

$$\frac{dy}{dx} = -ab \sin bx$$

Again, differentiating with respect to 'x'

$$d^2y/dx^2 = -ab^2 \cos bx = -b^2(a \cos bx) = -b^2y = d^2y/dx^2 + b^2y = 0$$

Hence proved

Ex 10.5

Question 1.

If F is the constant force generated by the motor of an automobile of mass M, its velocity V is given by $M \frac{dV}{dt} = F - kV$, where A is a constant. Express V in terms of t given that $V=0$ when $t=0$.

Solution:

Given equation $M \frac{dV}{dt} = F - kV$ (∵ F and k are constant)

$$M \frac{dV}{dt} = k(Fk - V) \int \frac{dV}{(Fk - V)} = \frac{kM}{k} \int \frac{dt}{Fk - V} = \log(Fk - V) = kMt + c$$

Given $V=0$ and $t=0$

$$\Rightarrow -\log Fk = c$$

Substituting in (1)

$$-\log(Fk - V) = ktM - \log(Fk) \log(Fk) - \log(F - V)k = ktM \log\left(\frac{Fk}{Fk - V}\right) = ktM \log\left(\frac{F}{F - V}\right) = e^{ktM} F = (F - V)k e^{ktM}$$

Question 2.

The velocity V, of a parachute falling vertically satisfies the equation,

$$V \frac{dV}{dx} = g(1 - V^2k^2)$$

where g and k are constants. If V and x are both initially zero, find V in terms of x.

Solution:

$$\text{Given } V \frac{dV}{dx} = g(1 - V^2k^2)$$

$$V \frac{dV}{dx} = g(k^2 - V^2k^2) \int \frac{V dV}{k^2 - V^2k^2} = g \int \frac{dx}{k^2 - V^2k^2} = \frac{1}{2} \log(k^2 - V^2k^2) = gk^2x + c \text{ given } v=0 \text{ and } x=0$$

$$-12 \log k^2 = c \Rightarrow \log 1k = c$$

Substitute in (1)

$$-12 \log(k^2 - V^2k^2) = gk^2x + \log 1k \log(1\sqrt{k^2 - V^2k^2}) - \log(1k) = gk^2x \log(k\sqrt{k^2 - V^2k^2}) = gk^2x \log\left(\frac{k}{k}\sqrt{k^2 - V^2k^2}\right)$$

$$-v_2 = e_{gx/k_2} k e^{-gx/k_2} = \sqrt{k_2 - v_2}$$

Squaring on both sides

$$k_2 e^{-2gx/k_2} = k_2 - v_2 \Rightarrow v_2 = k_2(1 - e^{-2gx/k_2})$$

Question 3.

Find the equation of the curve whose slope is $y - 1/x^2 + x$ and which passes through the point (1,0).

Solution:

Given that the slope is $y - 1/x^2 + x$

$$\text{i.e., } dy/dx = y - 1/x^2 + x \Rightarrow \int dy - 1/x^2 + x = \int dy - 1/x^2 + x \log(y-1) = \int (x+1) - x/(x+1) dx \log(y-1) = \int (1x - 1/x + 1) dx \log(y-1) = \log x - \log(x+1) + \log c \log(y-1) = \log(cxx+1) \Rightarrow y-1 = cxx+1$$

This passes through (1,0)

$$\Rightarrow -1 = c \Rightarrow c = -2$$

$\therefore$ Equation of the curve is $y-1 = -2xx+1$

$$y = 1 - 2xx + 1 = x + 1 - 2xx + 1 \Rightarrow y = 1 - xx + 1$$

Question 4.

Solve the following differential equations:

$$(i) dy/dx = \sqrt{1-y^2} \sqrt{1-x^2}$$

Solution:

$$dy/dx = \sqrt{1-y^2} \sqrt{1-x^2} \Rightarrow dy/\sqrt{1-y^2} = \sqrt{1-x^2} dx$$

Separating the variables

$$\int dy/\sqrt{1-y^2} = \int dx/\sqrt{1-x^2} \Rightarrow \sin^{-1}y = \sin^{-1}x + c$$

$$(ii) ydx + (1+x^2)\tan^{-1}xdy = 0$$

Solution:

$$ydx + (1+x^2)\tan^{-1}xdy = 0 \Rightarrow (1+x^2)\tan^{-1}xdy = -ydx$$

Separating the variables

$$\int dy/y = -\int dx/(1+x^2)\tan^{-1}x \Rightarrow \log y = -\log(\tan^{-1}x)$$

$$+ \log c \log y + \log(\tan^{-1}x) = \log c \log(y \tan^{-1}x) = \log c y \tan^{-1}x = c$$

$$(iii) \sin y dy/dx = a, y(0) = 1$$

Solution:

$$\sin y dy/dx = a, y(0) = 1 \Rightarrow \sin y dy = a dx \Rightarrow \int \sin y dy = \int a dx = x(\sin^{-1}a) + c$$

Given $y(0) = 1$

i.e., When $x=0, y=1$

$$\therefore 1 = 0(\sin^{-1}a) + c \Rightarrow c = 1$$

$$\therefore \text{The solution is } y = x \sin^{-1}a + 1 \Rightarrow y - 1 = x \sin^{-1}a \Rightarrow y - 1 = \sin^{-1}a \sin(y - 1) = a$$

$$(iv) dy/dx = e^{x+y} + x^3 e^y$$

Solution:

$$dy/dx = e^{x+y} + x^3 e^y \Rightarrow dy/dx = e^x \cdot e^y + x^3 e^y \Rightarrow dy/dx = e^y(e^x + x^3) \Rightarrow \int e^{-y} dy = \int (e^x + x^3) dx = e^x + x^4/4 + c$$

$$e^{-y} + e^{-y} x^4/4 = -c \Rightarrow e^{-y} + x^4/4 = c$$

$$(v) (e^y + 1) \cos x dx + e^y \sin x dy = 0$$

Solution:

$$(e^y + 1) \cos x dx + e^y \sin x dy = 0 \Rightarrow e^y \sin x dy = -(e^y + 1) \cos x dx$$

$$(e^y + 1) \cos x dx / e^y dy = - \int \cos x dx \sin x \log(e^y + 1) = -\log \sin x + \log c \log[(e^y + 1) + \log \sin x = \log c \log(e^y + 1) \sin x = \log c (e^y + 1) \sin x = c$$

$$(vi) (ydx - xdy) \cot(xy) = ny^2 dx$$

Solution:

$$(ydx - xdy) \cot(xy) = ny^2 dx$$

Dividing throughout by 'y²

$$(ydx - xdy)y^2 \cot(xy) = ndxd(xy) \cdot \cot(xy) = ndxf \cot(xy) \cdot d(xy) = n \int dx \log \sin(xy) = nx + c$$
$$\ln(xy) = e_{nx+c}$$

(vii) $dydx - x\sqrt{25-x^2} = 0$

Solution:

put $t = 25 - x^2$ $dt = -2xdxdt - 2 = xdx$

$$dydx - x\sqrt{25-x^2} = 0 \Rightarrow dydx = x\sqrt{25-x^2} \Rightarrow \int dy = \int x\sqrt{25-x^2} dx \Rightarrow \int dy = -12 \int \sqrt{t} dt = -12 \cdot \frac{2}{3} t^{3/2}$$
$$\frac{1}{2} + c \cdot 3y = -t^{3/2} + 3c \therefore 3y = -(25-x^2)^{3/2} + 3c$$

(viii) $x \cos y dy = e^x (x \log x + 1) dx$

Solution:

$$x \cos y dy = e^x (x \log x + 1) dx \Rightarrow x \cos y dy = e^x (x \log x + 1) dx \Rightarrow \int x \cos y dy = \int e^x (x \log x + 1) dx$$
$$x \log x dx + \int e^x \cdot 1 dx$$

Let us integrate $e^x \log x$ by using integration by parts

$$\int e^x \log x dx \text{ Let } u = \log x \Rightarrow du = \frac{1}{x} dx \Rightarrow \int u dv = uv - \int v du \Rightarrow \int e^x \log x dx = e^x \log x - \int e^x \cdot \frac{1}{x} dx$$

Substitute in (1) $\int x \cos y dy = \int e^x \sin y = e^x \log x dx + \int e^x dx \cdot \sin y = e^x \log x + c$

(ix) $\tan y dy dx = \cos(x+y) + \cos(x-y)$

Solution:

$$\tan y dy dx = \cos(x+y) + \cos(x-y)$$
$$+ \cos(x-y) \tan y dy dx = \cos x \cos y - \sin x \sin y + \cos x \cos y + \sin x \sin y \Rightarrow \tan y dy dx = 2 \cos x \cos y$$

y

Separating the variables

$$\int \tan y dy dx = 2 \int \cos x dx \Rightarrow \int \sec y \tan y dy = 2 \int \cos x dx \Rightarrow \sec y = 2 \sin x + c$$

(x) $dydx = \tan^2(x+y)$

Solution:

$$dydx = \tan^2(x+y)$$

Let $u = x+y$

Differentiating with respect to 'x'

$$du = 1 + dydx \Rightarrow du - 1 = dydx \text{ Now, } dydx = \tan^2(x+y) \Rightarrow du - 1 = \tan^2 u \Rightarrow du = 1 + \tan^2 u$$
$$du = \sec^2 u \Rightarrow \int du = \int \sec^2 u = \tan u \Rightarrow \tan u = x + c \Rightarrow \tan(x+y) = x + c$$
$$\Rightarrow x + c = \tan(x+y) \Rightarrow x + c = \frac{\sin(x+y)}{\cos(x+y)} \Rightarrow x + c \cos(x+y) = \sin(x+y)$$

Ex 10.6

Solve the following differential equations:

Question 1.

$$[x + y \cos(yx)] dx = x \cos(yx) dy$$

Solution:

$$[x + y \cos(yx)] dx = x \cos(yx) dy \Rightarrow dydx = \frac{x + y \cos(yx)}{x \cos(yx)}$$

This is a Homogeneous differential equation

Put $y = vx \Rightarrow dydx = v + x dvdx$ (1) $\Rightarrow v + x dvdx = \frac{x + vx \cos vx}{x \cos vx} = \frac{1 + v \cos v}{\cos v}$

$$x dvdx = \frac{1 + v \cos v}{\cos v} - v = \frac{1 + v \cos v - vx \cos v}{\cos v} = \frac{1 - v \sin v}{\cos v}$$

Separating the variables

$$\int \cos v \cdot dv = \int dx \cdot \sin v = \log x + c \Rightarrow \sin(yx) = \log x + c \Rightarrow \sin(yx) = \log x + c$$

$$\sin(yx) = \log |cx|$$

Question 2.

$$(x^3 + y^3)dy - x^2ydx = 0$$

Solution:

$$(x^3 + y^3)dy - x^2ydx = 0 \Rightarrow (x^3 + y^3)dy = x^2ydx \Rightarrow \frac{dy}{dx} = \frac{x^2y}{x^3 + y^3}$$

This is a Homogeneous differential equation.

$$\text{Put } y = vx \Rightarrow \frac{dy}{dx} = v + x \frac{dv}{dx} \quad (1) \Rightarrow v + x \frac{dv}{dx} = \frac{x^2(vx)}{x^3 + v^3x^3} = \frac{x^3v}{x^3(1 + v^3)} = \frac{v}{1 + v^3}$$

$$x \frac{dv}{dx} = \frac{v}{1 + v^3} - v = \frac{v - v(1 + v^3)}{1 + v^3} = \frac{v - v - v^4}{1 + v^3} = \frac{-v^4}{1 + v^3}$$

Separating the variables

$$\int (1 + v^3)v^4 dv = - \int \frac{dx}{x} \quad \int (1 + v^3)v^4 dv = - \int \frac{dx}{x} \quad \int (v^4 + v^7) dv = - \int \frac{dx}{x} \quad \frac{v^5}{5} + \frac{v^8}{8} = - \log x + \log c$$

$$-1x^3y^3 = \log(yx) \Rightarrow yx = e^{-x^3y^3} \therefore y = ce^{-x^3y^3}$$

Question 3.

$$ye^{xy}dx = (xe^{xy} + y)dy$$

Solution:

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$$ye^{xy}dx = (xe^{xy} + y)dy \Rightarrow \frac{dx}{dy} = \frac{xe^{xy} + y}{ye^{xy}} = \frac{x}{y} + \frac{1}{e^{xy}}$$

This is a Homogeneous differential equation

$$\text{Put } X = Vy \Rightarrow \frac{dx}{dy} = V + y \frac{dV}{dy}$$

$$(1) \Rightarrow V + y \frac{dV}{dy} = \frac{Vy}{y} + \frac{1}{e^{Vy}} = V + \frac{1}{e^{Vy}} \Rightarrow y \frac{dV}{dy} = \frac{1}{e^{Vy}} \Rightarrow \frac{dV}{dy} = \frac{1}{ye^{Vy}}$$

Separating the variables

$$\int e^{Vy} dy = \int \frac{dy}{y} \Rightarrow e^{Vy} = \log |cy| \quad \text{i.e., } e^{xy} = \log |cy| \quad [\because v = xy]$$

Question 4.

$$2xydx + (x^2 + 2y^2)dy = 0$$

Solution:

$$2xydx + (x^2 + 2y^2)dy = 0$$

$$(x^2 + 2y^2)dy = -2xydx$$

$$\frac{dx}{dy} = \frac{x^2 + 2y^2}{-2xy}$$

This is a Homogeneous differential equation

$$\text{Put } X = Vy \Rightarrow \frac{dx}{dy} = V + y \frac{dV}{dy}$$

$$(1) \Rightarrow V + y \frac{dV}{dy} = \frac{V^2y^2 + 2y^2}{-2Vy^2} = \frac{V^2 + 2}{-2V} \Rightarrow y \frac{dV}{dy} = \frac{V^2 + 2}{-2V} - V = \frac{V^2 + 2 - 2V^2}{-2V} = \frac{-V^2 + 2}{-2V}$$

Separating the variables

$$\int \frac{2V}{-V^2 + 2} dV = - \int \frac{dy}{y} \quad \int \frac{2V}{-V^2 + 2} dV = - \int \frac{dy}{y} \quad \text{Put } t = -V^2 + 2 \Rightarrow -2V dV = dt \Rightarrow \int \frac{-dt}{t} = - \log t + \log c$$

$$\log t^{1/3} = \log(cy) \Rightarrow t^{1/3} = cy \Rightarrow (-V^2 + 2)^{1/3} = cy \Rightarrow (-3x^2 + 2y^2)^{1/3} = cy \Rightarrow -3x^2 + 2y^2 = c^3y^3$$

Question 5.

$$(y^2 - 2xy)dx = (x^2 - 2xy)dy$$

Solution:

$$(y^2 - 2xy)dx = (x^2 - 2xy)dy \Rightarrow \frac{dx}{dy} = \frac{x^2 - 2xy}{y^2 - 2xy}$$

This is a Homogeneous differential equation

$$\text{Put } y = vx \Rightarrow \frac{dy}{dx} = v + x \frac{dv}{dx} \quad (1) \Rightarrow v + x \frac{dv}{dx} = \frac{x^2 - 2vx}{v^2x^2 - 2vx^2} = \frac{x^2(1 - 2v)}{x^2(v^2 - 2v)} = \frac{1 - 2v}{v^2 - 2v}$$

Separating the variables

$$\int \frac{v^2 - 2v}{1 - 2v} dv = \int \frac{dx}{x} \quad \int \frac{v^2 - 2v}{1 - 2v} dv = \int \frac{dx}{x} \quad \int \frac{v^2 - 2v + 1 - 1}{1 - 2v} dv = \int \frac{dx}{x} \quad \int \frac{v^2 - 2v + 1}{1 - 2v} dv - \int \frac{1}{1 - 2v} dv = \log x + \log c$$

Consider $\int 1-2v v^2-v dv$

Put $t=v^2-v$

$$dt=(2v-1)dv \Rightarrow dt=-(1-2v)dv \Rightarrow dt=(1-2v)dv$$

$$-13 \log t = \log x + \log c \Rightarrow -13 \log(v^2-v) = \log c x \Rightarrow \log(v^2-v) = \log c x \log(v^2-v)$$

$$-1/3 = \log c x (v^2-v) = (c x)^{1/3} (y^2 x^2 - y x) = c^{1/3} x^{1/3} x^3 (y^2 - x y x^2) = c x y^2 - x^2 y = c$$

Question 6.

$$x dy dx = y - x \cos^2(yx)$$

Solution:

$$x dy dx = y - x \cos^2(yx)$$

Dividing throughout by 'x'

$$dy dx = yx - \cos^2(yx)$$

This is a Homogeneous differential equation.

$$\text{Put } y=vx \Rightarrow dy dx = v + x dv dx \Rightarrow v + x \cdot dv dx = v - \cos^2 vx \cdot dv dx = v - \cos^2 v - vx \cdot dv dx = -c$$

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Separating the variables

$$\int \sec^2 v dv = - \int dx x \tan v = - \log x + \log \tan(yx) = \log(cx) \quad [\because v=yx] \Rightarrow cx = e^{\tan(yx)} \Rightarrow c = x e^{\tan(yx)}$$

Question 7.

$$(1+3e_{yx})dy + 3e_{yx}(1-yx)dx = 0, \text{ given that } y=0 \text{ when } x=1$$

Solution:

$$(1+3e_{yx})dy + 3e_{yx}(1-yx)dx = 0 \Rightarrow (1+3e_{yx})dy = -3e_{yx}(1-yx)dx \Rightarrow dy dx = -3e_{y/x}(1-yx) \Rightarrow 1+3e_{y/x}$$

This is a Homogeneous differential equation

$$\text{Put } y=vx \Rightarrow dy dx = v + x dv dx \Rightarrow v + x \cdot dv dx = -3e_v(1-v) \Rightarrow 1+3e_v x dv dx = -3e_v + 3ve_v \Rightarrow 1+3e_v$$

$$e_v - vx dv dx = -3e_v + 3ve_v - v - 3ve_v \Rightarrow 1+3e_v$$

$$x dv dx = -(3e_v + v) \Rightarrow 1+3e_v$$

Separating the variables

$$\int (1+3e_v) dv \Rightarrow 3e_v + v = - \int dx x \Rightarrow \text{Put } t=3e_v + v \Rightarrow dt = (3e_v + 1) dv \Rightarrow \int dt t = - \int dx x$$

$$\log t = - \log x + \log c \Rightarrow \log t = \log(cx) \Rightarrow t = cx \Rightarrow 3e_{y/x} + yx = c \Rightarrow c = x(3e_{y/x} + yx)$$

Given $y=0$ when $x=1$

$$1(3e_0 + 0) = c \Rightarrow c = 3$$

$$\therefore \text{The solution is } x(3e_{y/x} + yx) = 3$$

$$\Rightarrow 3x e_{y/x} + y = 3$$

Question 8.

$(x^2+y^2)dy = xy dx$. It is given that $y(1)=1$ and $y(x_0)=e$. Find the value of x_0 .

Solution:

$$(x^2+y^2)dy = xy dx \Rightarrow x^2 dy + y^2 dy = xy dx \Rightarrow x^2 dy = xy dx - y^2 dy$$

This is a Homogeneous differential equation

$$\text{Put } x=vy \Rightarrow dx dy = v + y \cdot dv dy \Rightarrow v + y \cdot dv dy = v^2 y^2 + y^2 v y^2 = y^2(v^2 + 1) \Rightarrow v y^2 dv dy = v^2 + 1 \Rightarrow v$$

$$v y dv dy = v^2 + 1 \Rightarrow v^2 v = 1 \Rightarrow v$$

Separating the variables

$$\int v dv = \int dy y v^2 \Rightarrow \log y + \log c x^2 y^2 = \log c y \Rightarrow c y = e^{x^2/2 y^2}$$

Given $y(1)=1$

$$\text{i.e., When } x=1, y=1$$

$$\text{From (2)} \Rightarrow c = e_{12}$$

$$\therefore (2) \Rightarrow e_{12} y = e^{x^2/2 y^2}$$

Now, given $y(x_0)=e$

i.e., When $x=x_0, y=e$

$$\therefore e^{12} \cdot e = e^{x_{20}/2} e^{2e^{3/2}} = e^{x_{20}/2} e^{2 \cdot 3} = x_{20} 2 e^2 \therefore x_{20} = 3 e^2 x_0 = \pm \sqrt{3} e$$

Ex 10.7

Solve the following Linear differential equations:

Question 1.

$$\cos x dy + y \sin x = 1$$

Solution:

$$\cos x dy + y \sin x = 1$$

Dividing throughout by ' $\cos x$ '

$$dy + y \tan x = \sec x$$

This is a linear differential equation of the type $dy + Py = Q$ Where $P = \tan x$, $Q = \sec x$

$$\int P dx = \int \tan x dx = \log \sec x \text{ I.F. } = e^{\int P dx} = e^{\log \sec x} = \sec x \text{ Solution}$$

$$\text{is } y e^{\int P dx} = \int Q e^{\int P dx} dx + c y \sec x = \int \sec x \cdot \sec x \cdot dx + c y \sec x = \int \sec^2 x dx + c y \sec x = \tan x + c (\div \sec x) y = \sin x + c \cos x$$

Question 2.

$$(1-x^2) dy + xy = 1$$

Solution:

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$$(1-x^2) dy + xy = 1$$

Dividing throughout by $(1-x^2)$

$$dy + \frac{xy}{1-x^2} = \frac{1}{1-x^2}$$

This is a linear differential equation of the form $dy + Py = Q$

Where $P = -\frac{x}{1-x^2}$ and $Q = \frac{1}{1-x^2}$

$$\int P dx = \int \frac{-x}{1-x^2} dx = \log(1-x^2) \text{ I.F. } = e^{\int P dx} = e^{\log(1-x^2)} = (1-x^2) \text{ Solution}$$

$$\text{is } y e^{\int P dx} = \int Q e^{\int P dx} dx + c y \sqrt{1-x^2} = \int \frac{1}{1-x^2} \sqrt{1-x^2} dx + c y \sqrt{1-x^2} = \int \frac{1}{\sqrt{1-x^2}} dx + c y \sqrt{1-x^2}$$
$$\sqrt{1-x^2} = \sin^{-1} x + c (\div \sqrt{1-x^2}) y = \sin^{-1} x \sqrt{1-x^2} + c (1-x^2)^{-1/2}$$

Question 3.

$$dy + yx = \sin x$$

Solution:

$$dy + yx = \sin x$$

This is a linear differential equation of the form $dy + Py = Q$

Where $P = x$ and $Q = \sin x$ $\int P dx = \int x dx = \frac{x^2}{2}$ I.F. $= e^{\int P dx} = e^{x^2/2} = x$ Solution

$$\text{is } y e^{\int P dx} = \int Q e^{\int P dx} dx + c \text{ Consider } \int x \sin x dx = -x \cos x + \sin x + c x y + x \cos x = \sin x + c x (y + \cos x) = \sin x + c$$

Question 4.

$$(x^2+1) dy + 2xy = \sqrt{x^2+4}$$

Solution:

$$(x^2+1) dy + 2xy = \sqrt{x^2+4}$$

Dividing throughout by (x^2+1)

$$dy + \frac{2xy}{x^2+1} = \frac{\sqrt{x^2+4}}{x^2+1}$$

This is a Linear differential equation of the form $dy + Py = Q$ where $P = \frac{2x}{x^2+1}$ and $Q = \frac{\sqrt{x^2+4}}{x^2+1}$

$\int P dx = \int 2x + \frac{1}{x^2} dx = \log(1+x^2)$ [$\because \int f(x)f'(x) dx = \log f(x)$] I.F. = $e^{\int P dx} = e^{\log(1+x^2)} = 1+x^2$ Solution

is $y e^{\int P dx} = \int Q e^{\int P dx} dx + c$ $y(1+x^2) = \int \sqrt{x^2+4} \cdot 2x dx + c$ $y(1+x^2) = \int \sqrt{x^2+4} dx + c$ $\therefore y(1+x^2) = x^2 \sqrt{x^2+4} + 2 \log(x + \sqrt{x^2+4}) + c$

Question 5.

$$(2x - 10y^3) dy + y dx = 0$$

Solution:

$$(2x - 10y^3) dy + y dx = 0 \quad y dx = (10y^3 - 2x) dy \quad y dx dy = 10y^3 dy - 2x dy \quad 2xy = 10y^2$$

This is a Linear differential equation of the form $dx dy + P x = Q$ where $P = 2y$ and $Q = 10y^2$

$$\int P dy = \int 2y dy = y^2 = \log y^2 \quad \text{I.F.} = e^{\int P dy} = e^{\log y^2} = y^2$$

Solution is $x e^{\int P dy} = \int Q e^{\int P dy} dy + c$

$$x y^2 = \int 10 y^2 (y^2) dy + c \quad x y^2 = 10 \int y^4 dy + c \quad x y^2 = 10 y^5 + c \Rightarrow x y^2 = 2 y^5 + c$$

Question 6.

$$x \sin x dy + (x \cos x + \sin x) y = \sin x$$

Solution:

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$$x \sin x dy + (x \cos x + \sin x) y = \sin x$$

Dividing throughout by ' $x \sin x$ '

$$dy + (\cot x + \frac{1}{x}) y = \frac{1}{x}$$

This is a Linear differential equation of the type $dy + P y = Q$

where $P = \cot x + \frac{1}{x}$ and $Q = \frac{1}{x}$ $\int P dx = \int (\cot x + \frac{1}{x}) dx = \log \sin x + \log x = \log(x \sin x)$ I.F. = $e^{\int P dx} = e^{\log(x \sin x)} = x \sin x$

Solution is $y e^{\int P dx} = \int Q e^{\int P dx} dx + c$

$$y(x \sin x) = \int \frac{1}{x} \cdot x \sin x dx + c \quad y x \sin x = \int \sin x dx + c \quad y x \sin x = -\cos x + c \quad y x \sin x = \cos x + c$$

Question 7.

$$(y - e^{\sin^{-1} x}) dx + \sqrt{1-x^2} dy = 0$$

Solution:

$$(y - e^{\sin^{-1} x}) dx + \sqrt{1-x^2} dy = 0 \quad \sqrt{1-x^2} dy = - (y - e^{\sin^{-1} x}) dx \quad \frac{dy}{dx} = - \frac{y - e^{\sin^{-1} x}}{\sqrt{1-x^2}}$$

$$(y - e^{\sin^{-1} x}) \sqrt{1-x^2} \frac{dy}{dx} + y \sqrt{1-x^2} = e^{\sin^{-1} x} \sqrt{1-x^2}$$

This is a Linear differential equation of the type $dy + P y = Q$

where $P = \frac{1}{\sqrt{1-x^2}}$ and $Q = e^{\sin^{-1} x} \sqrt{1-x^2}$ $\int P dx = \int \frac{1}{\sqrt{1-x^2}} dx = \sin^{-1} x$ I.F. = $e^{\int P dx} = e^{\sin^{-1} x}$

Solution is $y e^{\int P dx} = \int Q e^{\int P dx} dx + c$

$$y e^{\sin^{-1} x} = \int e^{\sin^{-1} x} \sqrt{1-x^2} e^{\sin^{-1} x} dx + c \quad y e^{\sin^{-1} x} = \int (e^{\sin^{-1} x})^2 \sqrt{1-x^2} dx + c \quad \text{Consider } \int (e^{\sin^{-1} x})^2 \sqrt{1-x^2} dx$$

Put $t = \sin^{-1} x$

$$dt = \frac{1}{\sqrt{1-x^2}} dx \therefore \int (e^{\sin^{-1} x})^2 \sqrt{1-x^2} dx = \int (e^t)^2 dt = \int e^{2t} dt = \frac{e^{2t}}{2} = \frac{e^{2 \sin^{-1} x}}{2}$$

Substituting in (1) $\Rightarrow y e^{\sin^{-1} x} = \frac{e^{2 \sin^{-1} x}}{2} + c$

Question 8.

$$dy + y(1-x)\sqrt{x} = 1 - \sqrt{x}$$

Solution:

$$y \frac{dy}{dx} + y(1-x)\sqrt{x} = 1 - \sqrt{x}$$

This is a Linear differential equation of the type $y \frac{dy}{dx} + Py = Q$ where $P = 1(1-x)\sqrt{x}$ and $Q = 1 - \sqrt{x}$

$$\text{Let } u = \sqrt{x} \Rightarrow du = \frac{1}{2\sqrt{x}} dx$$

$$2du = \frac{dx}{\sqrt{x}} \therefore \int 1(1-x)\sqrt{x} dx = 2 \int du (1-u^2) = 2 \left[\log |u+1| - \log |u-1| \right]$$

$$(u-1) \left| \frac{1}{u+1} \right| = \log \left| \frac{1}{(u+1)(u-1)} \right| = \log \left| \frac{1}{\sqrt{x}+1\sqrt{x}-1} \right| \text{ I.F.}$$

$$e^{\int P dx} = e^{\log \left| \frac{1}{\sqrt{x}+1\sqrt{x}-1} \right|} = \left| \frac{1}{\sqrt{x}+1\sqrt{x}-1} \right|$$

Solution is $y e^{\int P dx} = \int Q e^{\int P dx} dx + c$

$$y \left| \frac{1}{\sqrt{x}+1\sqrt{x}-1} \right| = \int \left| \frac{1}{\sqrt{x}+1\sqrt{x}-1} \right| (1-\sqrt{x}) dx + c y (1+\sqrt{x}-\sqrt{x}) = \int (\sqrt{x}+1) dx +$$

$$c = 23x^{3/2} + x + c \therefore y(1+\sqrt{x}-\sqrt{x}) = 23x\sqrt{x} + x + c$$

Question 9.

$$(1+x+xy^2) \frac{dy}{dx} + (y+y^3) = 0$$

Solution:

$$(1+x+xy^2) \frac{dy}{dx} + (y+y^3) = 0 \Rightarrow (1+x+xy^2) \frac{dy}{dx} = -$$

$$(y+y^3) \frac{dx}{dy} = -(1+x+xy^2) y + y^3 \frac{dx}{dy} = -1y(1+y^2) - x(1+y^2) \frac{dx}{dy} + xy = -1/y(1+y^2)$$

This is a Linear differential equation of the type $\frac{dx}{dy} + Px = Q$

where $P = 1/y$ and $Q = -1/y(1+y^2)$ $\int P dy = \int 1/y dy = \log y$ I.F. $= e^{\int P dy} = e^{\log y} = y$

Solution is $x e^{\int P dy} = \int Q e^{\int P dy} dy + c$

$$xy = - \int 1/y(1+y^2) y dy + c$$

$$xy = - \int 1/(1+y^2) dy + c xy = -\tan^{-1}y + c xy + \tan^{-1}y = c$$

Question 10.

$$\frac{dy}{dx} + yx \log x = \sin 2x \log x$$

Solution:

$$\frac{dy}{dx} + yx \log x = \sin 2x \log x$$

This is a Linear differential equation of the type $\frac{dy}{dx} + Py = Q$

where $P = 1x \log x$ and $Q = \sin 2x \log x$

$$e^{\int P dx} = \int 1x \log x dx = \log(\log x) \left[\because \int f(x)f'(x) dx = \log f(x) \right] \text{ I.F. } = e^{\int P dx} = e^{\log(\log x)} = \log x$$

Solution is $y e^{\int P dx} = \int Q e^{\int P dx} dx + c$

$$y \log x = \int \sin 2x \log x \log x dx + c y \log x = \int \sin 2x dx + c$$

$$y \log x = -\cos 2x/2 + c y \log x + \cos 2x/2 = c$$

Question 11.

$$(x+a) \frac{dy}{dx} - 2y = (x+a)^4$$

Solution:

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$$(x+a) \frac{dy}{dx} - 2y = (x+a)^4$$

Dividing throughout by $(x+a)$

$$\frac{dy}{dx} - 2y/(x+a) = (x+a)^3$$

This is a Linear differential equation of the type $\frac{dy}{dx} + Py = Q$ where $P = -2/(x+a)$ and $Q = (x+a)^3$

$$\int P dx = -2 \int \frac{dx}{x+a} = -2 \log(x+a) = \log 1/(x+a)^2 \text{ I.F. } = e^{\int P dx} = e^{\log 1/(x+a)^2} = 1/(x+a)^2$$

Solution is $y e^{\int P dx} = \int Q e^{\int P dx} dx + c$

$$y(1+x+a)^2 = f(x+a)^3 \frac{1}{(x+a)^2} dx + cy(1+x+a)^2 = f(x+a) dx + cy(1+x+a)^2 = (x+a)^2 + c^2 y = (x+a)^4 + 2c(x+a)^2$$

Question 12.

$$y \frac{dy}{dx} = \sin 2x^{1+x^3} - 3x^{2+1+x^3} y$$

Solution:

$$y \frac{dy}{dx} = \sin 2x^{1+x^3} - 3x^{2+1+x^3} y \quad y \frac{dy}{dx} + 3x^{2+1+x^3} y = \sin 2x^{1+x^3}$$

This is a Linear differential equation of the type $\frac{dy}{dx} + Py = Q$

where $P = 3x^{2+1+x^3}$ and $Q = \sin 2x^{1+x^3}$

$$\int P dx = \int 3x^{2+1+x^3} dx = \log(1+x^3) \quad [\because \int f(x)f'(x) dx = \log f(x)] \quad \text{I.F.} = e^{\int P dx} = e^{\log(1+x^3)} = 1+x^3$$

Solution is $y e^{\int P dx} = \int Q e^{\int P dx} dx + c$

$$y(1+x^3) = \int \sin 2x^{1+x^3} (1+x^3) dx + cy(1+x^3) = \int \sin 2x dx + cy(1+x^3) = \int (1 - \cos 2x^2) dx + cy(1+x^3) = 12[x - \sin 2x^2] + cy(1+x^3) = x^2 - \sin 2x^4 + c$$

Question 13.

$$x \frac{dy}{dx} + y = x \log x$$

Solution:

$$x \frac{dy}{dx} + y = x \log x$$

Dividing throughout by 'x'

$$\frac{dy}{dx} + yx = \log x$$

This is a Linear differential equation of the type $\frac{dy}{dx} + Py = Q$

where $P = \frac{1}{x}$ and $Q = \log x$

$$\int P dx = \int \frac{1}{x} dx = \log x \quad \text{I.F.} = e^{\int P dx} = e^{\log x} = x$$

Solution is

$$y e^{\int P dx} = \int Q e^{\int P dx} dx + cyx = \int x \log x dx + c$$

Now, Using integration by parts method to integrate $\int x \log x dx$

$$\text{Let } u = \log x \quad dv = x dx$$

$$du = \frac{1}{x} dx \quad v = x^2 \quad \int u dv = uv - \int v du \quad \int x \log x dx = x^2 \log x - \int x^2 \frac{1}{x} dx = x^2 \log x - \frac{x^2}{2} + c = x^2 \left[\log x - \frac{1}{2} \right] + c$$

$$\therefore \text{The solution is } xy = x^2 \left[\log x - \frac{1}{2} \right] + c \quad (\text{or}) \quad 4xy = 2x^2 \log x - x^2 + 4c$$

Question 14.

$$x \frac{dy}{dx} + 2y - x^2 \log x = 0$$

Solution:

$$x \frac{dy}{dx} + 2y - x^2 \log x = 0 \quad x \frac{dy}{dx} + 2y = x^2 \log x \quad \frac{dy}{dx} + 2yx = x \log x$$

This is a Linear differential equation of the type $\frac{dy}{dx} + Py = Q$

where $P = \frac{2}{x}$ and $Q = x \log x$ $\int P dx = 2 \int \frac{1}{x} dx = 2 \log x = \log x^2$ $\text{I.F.} = e^{\int P dx} = e^{\log x^2} = x^2$

Solution is $y e^{\int P dx} = \int Q e^{\int P dx} dx + c$

$$y(x^2) = \int (x \log x) x^2 dx + cy(x^2) = \int x^3 \log x dx + c$$

Using integration by parts method

$$\text{Let } u = \log x \quad dv = x^3 dx$$

$$du = \frac{1}{x} dx \quad v = \frac{x^4}{4} \quad \int u dv = uv - \int v du \quad \int x^3 \log x dx = \frac{x^4}{4} \log x - \int \frac{x^4}{4} \frac{1}{x} dx = \frac{x^4}{4} \log x - \frac{x^4}{16} + c$$

$$\therefore \text{The solution is } yx^2 = \frac{x^4}{4} \log x - \frac{x^4}{16} + c$$

Question 15.

$$\frac{dy}{dx} + 3yx = 1x^2, \text{ given that } y=2 \text{ when } x=1$$

Solution:

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$$dydx + 3yx = 1x^2$$

This is a Linear differential equation of the type $dydx + Py = Q$ where $P = 3x$ and $Q = 1x^2$

$$\int Pdx = \int 3x dx = 3 \log x = \log x^3 \text{ I.F. } = e^{\int Pdx} = e^{\log x^3} = x^3$$

$$\text{Solution is } ye^{\int Pdx} = \int Qe^{\int Pdx} dx + c$$

$$yx^3 = \int 1x^2 x^3 dx + cyx^3 = \int x dx + cyx^3 = x^2/2 + c$$

Given $y=2$ when $x=1$

$$(2)(1)^3 = 1/2 + c \Rightarrow c = 2 - 1/2 = 3/2$$

$\therefore$ The solution is $yx^3 = x^2/2 + 3/2$

$$2x^3y = x^2 + 3$$

Ex 10.8

Question 1.

The rate of increase in the number of bacteria in a certain bacteria culture is proportional to the number present. Given that the number triples in 5 hours, find how many bacteria will be present after 10 hours?

Solution:

Let 'P' be the number of bacteria present at time 't'

$$\text{Given } dP/dt \propto P$$

$$\Rightarrow dP/dt = kP \int dP/P = \int k dt \log P = kt + c \Rightarrow P = e^{kt+c} = e^{kt} \cdot e^c = e^{kt} \cdot P_0 \dots (1)$$

Initially, when $t=0$, let $P=P_0$

$$P_0 = e^c \cdot P_0 \Rightarrow c = 0$$

$$\text{Given } t=5, P=3P_0$$

Substituting in (1)

$$3P_0 = e^{5k} \cdot P_0 \Rightarrow e^{5k} = 3$$

Again, when $t=10$

$$P = e^{10k} \cdot P_0 = (e^{5k})^2 \cdot P_0 = (3)^2 P_0 = 9P_0$$

[$\because k$ is constant]

After 10 hours, the number of bacteria is 9 times the original bacteria present.

Question 2.

Find the population of a city at any time t , given that the rate of increase of population is proportional to the population at that instant and that in a period of 40 years the population increased from 3,00,000 to 4,00,000.

Solution:

Let the population of a city be 'P' at time 't'

$$\text{Given } dP/dt \propto P$$

$$\Rightarrow dP/dt = kP \int dP/P = \int k dt \log P = kt + c \Rightarrow P = e^{kt+c} = e^{kt} \cdot e^c$$

$$\text{Given, when } t=0, P=3,00,000$$

Substituting in (1)

$$3,00,000 = e^c \cdot P_0 \therefore (1) \Rightarrow P = 300000 e^{kt}$$

$$\text{Again, given when } t=40, P=4,00,000$$

$$\therefore 400000 = 300000 e^{40k} \Rightarrow e^{40k} = 4/3 \Rightarrow 40k = \log(4/3) \Rightarrow k = \frac{1}{40} \log(4/3)$$

$$\therefore \text{The population is } P = 300000 e^{\log(4/3) \cdot t/40}$$

$$P = 300000 (4/3)^{t/40}$$

Question 3.

The equation of electromotive force for an electric circuit containing resistance and self

inductance is $E = Ri + L \frac{di}{dt}$, where E is the electromotive force is given to the circuit, R the resistance and L , the coefficient of induction. Find the current i at time t when $E = 0$.

Solution:

Given equation of electromotive force is

$$E = Ri + L \frac{di}{dt}$$

$$\text{Let } E = E_0$$

$$\therefore \text{Equation is } L \frac{di}{dt} + Ri = E_0 \Rightarrow L \frac{di}{dt} + Ri = E_0$$

This is a Linear differential equation

$$P(t) = RL \text{ and } Q(t) = E_0 \Rightarrow \int P(t) dt = RL \int dt = Rt \quad I.F. = e^{\int P(t) dt} = e^{Rt/L}$$

Now, solution is $i e^{Rt/L} = E_0 \int e^{Rt/L} dt + C$

$$i e^{Rt/L} = E_0 \cdot \frac{e^{Rt/L}}{R/L} + C \Rightarrow i e^{Rt/L} = E_0 R e^{Rt/L} + C$$

Given $t = 0$ and $i = 0$

$$0 = E_0 R + C \Rightarrow C = -E_0 R$$

Substituting in (1)

$$i = E_0 R e^{Rt/L} - E_0 R = E_0 R (e^{Rt/L} - 1)$$

Question 4.

The engine of a motor boat moving at 10 m/s is shut off. Given that the retardation at any subsequent time (after shutting off the engine) equal to the velocity at that time. Find the velocity after 2 seconds of switching off the engine.

Solution:

Given, the retardation experienced by a moving motor boat after its engine is shut off is given by

$$\frac{dv}{dt} = -v \Rightarrow \int \frac{dv}{v} = - \int dt \Rightarrow \log v = -t + C \Rightarrow v = e^{-t+C} = e^C \cdot e^{-t}$$

Given, $v = 10$, when $t = 0$

$$\text{Now (1) } \therefore C = 10 \Rightarrow v = 10e^{-t}$$

$$\text{Again, given } t = 2 \Rightarrow v = 10e^{-2}$$

Question 5.

Suppose a person deposits 10,000 Indian rupees in a bank account at the rate of 5% per annum compounded continuously. How much money will be in his bank account 18 months later?

Solution:

Let P be the principal

Rate of interest 5%

$$\Rightarrow \frac{dP}{dt} = 0.05P \Rightarrow \int \frac{dP}{P} = \int 0.05 dt \Rightarrow \log P = 0.05t + C \Rightarrow P = e^{0.05t+C} = e^C \cdot e^{0.05t}$$

Given, when $t = 0$, $P = 10000$

Question 6.

Assume that the rate at which radioactive nuclei decay is proportional to the number of such nuclei that are present in a given sample. In a certain sample 10% of the original number of radioactive nuclei have undergone disintegration in a period of 100 years. What percentage of the original radioactive nuclei will remain after 1000 years?

Solution:

Let $x(t)$ denote the number of nuclei present in a sample at anytime ' t '

Given $\frac{dx}{dt} = kx$ [$\because k$ is a negative constant]

$$\int \frac{dx}{x} = \int k dt \Rightarrow \log x = kt + C$$

Let x_0 denote the number of nuclei present in the given sample at time $t = 0$

$$\therefore \text{from (1) } \Rightarrow C = \log x_0$$

$$\therefore x = x_0 e^{kt}$$

Given that when $t = 100$ years, $x = x_0 - 10\% x_0 = 910 x_0$

$\therefore$ Required percentage after 1000 years

$=X_0 \times 100 = X_0 e^{1000k} X_0 \times 100 = (e^{1000k})_{10} \times 100 [\because e^{1000k} = 910] = (910)_{10} \times 100 = 910108 \therefore 910108$
 % of radioactive element will remain after 1000 years.

Question 7.

Water at temperature 100°C cools in 10 minutes to 80°C in a room temperature of 25°C . Find

(i) The temperature of water after 20 minutes

(ii) The time when the temperature is 40°C

$$[\log_e 1115 = -0.3101; \log_e 5 = 1.6094]$$

Solution:

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Let 'T' be the temperature in time 't'

$$\text{Given } \frac{dT}{dt} \propto (T-25) \Rightarrow \frac{dT}{T-25} = -k(T-25), k > 0 \int \frac{dT}{T-25} = -k \int dt \log(T-25) = -kt + c$$

Given when $t=0$, $T=100^\circ\text{C}$

$$\Rightarrow c = \log 75(1) \Rightarrow \log(T-25) = -kt + \log 75 \log(T-25) = -kt \quad kt = \log(75/T-25)$$

Again, given $t=10$, $T=80$

$$\Rightarrow 10k = \log(75/55) \quad 10k = \log(15/11) \therefore k = \frac{1}{10} \log(15/11) = \frac{1}{10}(0.3101) \quad k = 0.03101 \therefore 0.03101$$

$$t = \log(75/T-25)$$

(i) When $t=20$, $T = ?$

$$0.6202 = \log(75/T-25) \quad (75/T-25) = e^{0.6202} = 1.8593$$

$$T-25 = 40.38 \quad T = 65.38^\circ\text{C}$$

(ii) When $T=48$, $t = ?$

$$\Rightarrow 0.03101t = \log(75/15) \quad 0.03101t = \log 5 \quad 0.03101t = 1.6094 \quad t = \frac{1.6094}{0.03101} = 52 \text{ minutes.}$$

(i) When $t=20$, $T = ?$

$$0.6202 = \log(75/T-25) \quad (75/T-25) = e^{0.6202} = 1.8593$$

$$T-25 = 40.38 \quad T = 65.38^\circ\text{C}$$

(ii) When $T=48$, $t = ?$

$$\Rightarrow 0.03101t = \log(75/15) \quad 0.03101t = \log 5 \quad 0.03101t = 1.6094 \quad t = \frac{1.6094}{0.03101} = 52 \text{ minutes.}$$

Question 8.

At 10.00 A.M. a woman took a cup of hot instant coffee from her microwave oven placed it on a nearby Kitchen counter to cool. At this instant the temperature of the coffee was 180°F , and 10 minutes later it was 160°F . Assume that constant temperature of the kitchen was 70°F .

(i) What was the temperature of the coffee at 10.15 A.M.?

(ii) The woman likes to drink coffee when its temperature is between 130°F and 140°F . between what times should she have drunk the coffee?

Solution:

Change in Temperature is proportional to the difference in temperature (Newton's Cooling Law)

$$\therefore \frac{dT}{dt} \propto (70-T) \quad \frac{dT}{T-70} = -k(T-70) \int \frac{dT}{T-70} = -k \int dt -\log(T-70) = kt + c \therefore T = 70 + ce^{-kt} \text{ Given } T(0) = 180 \Rightarrow 180 = 70 + c \quad c = 110 \text{ and } T(10) = 160 \therefore 160 = 70 + 110e^{-10k} \quad e^{-10k} = \frac{90}{110} = \frac{9}{11}$$

$$160 - 70 = 110e^{-10k} \quad 90 = 110e^{-10k} \quad 9 = 11e^{-10k} \quad -10k = \log \frac{9}{11} \quad k = \frac{1}{10} \log \frac{11}{9}$$

(i) When $t=15$ min

$$\text{then } T(15) = 70 + 110e^{-15k} = 70 + 110e^{-15 \cdot \frac{1}{10} \log \frac{11}{9}} = 70 + 110e^{-\frac{3}{2} \log \frac{11}{9}} = 70 + 110e^{\log \frac{(911)^{3/2}}{110}} = 70 + 110(911)^{3/2} = 70 + 81.39 = 151.39$$

(ii) Substituting $T=130$ and $T=140$

We get $t=22.523$

Question 9.

A pot of boiling water at 100°C is removed from a stove at time $t=0$ and left to cool in the kitchen. After 5 minutes, the water temperature has decreased to 80°C , and another 5 minutes later it has dropped to 65°C . Determine the temperature of the kitchen.

Solution:

Let A be the temperature of the kitchen

Given, $dT/dt \propto (A-T)$ [$\because$ Newton's cooling

law] $dT/dt = k(A-T)$ $\int dT/(A-T) = \int k dt$ $\log(A-T) = kt + c \Rightarrow T = A + ce^{-kt} \dots (1)$

Given, $T(0)=100$,

$T(5)=80$, $T(10)=65$

Substituting in (1), we get

$$100 = A + c \quad 80 = A + ce^{-5k} \quad 65 = A + ce^{-10k}$$

Solving these linear equations,

We get $A=11^{\circ}$ (Kitchen Temperature)

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Question 10.

A tank initially contains 50 liters of pure water. Starting at time $t=0$ a brine containing with 2 grams of dissolved salt per litre flows into the tank at the rate of 3 liters per minute. The mixture is kept uniform by stirring and the well-stirred mixture simultaneously flows out of the tank at the same rate. Find the amount of salt present in the tank at any time $t>0$.

Solution:

Let ' x ' be the amount of salt present

$$dx/dt = c(t)r(t) - x(t)v(t) \cdot r_o(t) \quad dv/dt = r(t) - r_o(t)$$

Given, $v(0)=50$, $c(t)=2$, $r(t)=3$ and $r_o(t)=3$

$$(2) \Rightarrow dv/dt = 3 - 3 = 0 \quad \int dv = 0 + c \Rightarrow v = cv(0) = 50 \Rightarrow c = 50$$

$$(1) \Rightarrow dx/dt = (2)$$

$$(3) \Rightarrow x50(3) dx/dt = 3(100 - x50) = -350[x - 100] \quad \int dx/(x - 100) = -350 \int dt \log(x - 100) = -350t + cx - 100 = e^{-350t} \cdot cx = 100 + 50e^{-350t} = 50(2 + e^{-350t})$$

$$0t + cx - 100 = e^{-350t} \cdot cx = 100 + 50e^{-350t} = 50(2 + e^{-350t})$$

Since the tank initially contains only pure water, the initial amount of salt is zero. i.e., $x(0)=0$

Ex 10.9

Choose the correct or the most suitable answer from the given four alternatives:

Question 1.

The order and degree of the differential equation $d^2y/dx^2 + (dy/dx)^{1/3} + x^{1/4} = 0$ are respectively.

(a) 2,3

(b) 3,3

(c) 2,6

(d) 2,4

Solution:

(a) 2,3

Hint:

$$d^2y/dx^2 + (dy/dx)^{1/3} + x^{1/4} = 0 \quad d^2y/dx^2 + x^{1/4} = -(dy/dx)^{1/3}$$

Raising the power both sides to ' 3 '

$$[d^2y/dx^2 + x^{1/4}]^3 = -(dy/dx)$$

Order =2,

degree =3

Question 2.

The differential equation representing the family of curves $y = A \cos(x+B)$, where A and B are parameters, is

- (a) $d^2y/dx^2 - y = 0$
- (b) $d^2y/dx^2 + y = 0$
- (c) $d^2y/dx^2 = 0$
- (d) $d^2x/dy^2 = 0$

Solution:

- (b) $d^2y/dx^2 + y = 0$

Hint:

$$y = A \cos(x+B) \Rightarrow dy/dx = -A \sin(x+B) \Rightarrow d^2y/dx^2 = -A \cos(x+B) = -y \Rightarrow d^2y/dx^2 + y = 0$$

Question 3.

The order and degree of the differential equation $\sqrt{\sin x}(dx+dy) = \sqrt{\cos x}(dx-dy)$ is

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- (a) 1,2
- (b) 2,2
- (c) 1,1
- (d) 2,1

Solution:

- (c) 1, 1

Hint:

$$\sqrt{\sin x}(dx+dy) = \sqrt{\cos x}(dx-dy)$$

Since, the first order derivatives are involved, Order = 1 and degree 1 .

Question 4.

The order of the differential equation of all circles with centre at (h,k) and radius ' a ' is

- (a) 2
- (b) 3
- (c) 4
- (d) 1

Solution:

- (a) 2

Hint:

Equation of circle is $(x-h)^2 + (y-k)^2 = a^2$

Equation is to be differentiated twice as two parameters are given.

$\therefore$ Order = 2

Question 5.

The differential equation of the family of curves $y = Ae^x + Be^{-x}$, where A and B are arbitrary constants is

- (a) $d^2y/dx^2 + y = 0$
- (b) $d^2y/dx^2 - y = 0$
- (c) $dy/dx + y = 0$
- (d) $dy/dx - y = 0$

Solution:

- (b) $d^2y/dx^2 - y = 0$

Hint:

$$y = Ae^x + Be^{-x} \Rightarrow dy/dx = Ae^x - Be^{-x} \Rightarrow d^2y/dx^2 = Ae^x + Be^{-x} = y \Rightarrow d^2y/dx^2 - y = 0$$

Question 6.

The general solution of the differential equation $dy/dx = yx$ is

- (a) $xy=k$
- (b) $y=k\log x$
- (c) $y=kx$
- (d) $\log y=kx$

Solution:

- (c) $y=kx$

Question 7.

The solution of the differential equation (a) straight lines $2x \frac{dy}{dx} - y = 3$

- (b) circles
- (c) parabola
- (d) ellipse

Solution:

- (c) parabola Hint:

$$2x \frac{dy}{dx} - y = 3 \Rightarrow 2x \frac{dy}{dx} = y + 3 \Rightarrow 2 \int \frac{dy}{y+3} = \int \frac{dx}{x} \Rightarrow 2 \log(y+3) = \log x + \log c \Rightarrow \log(y+3)^2 = \log cx$$

Question 8.

The solution of $\frac{dy}{dx} + p(x)y = 0$ is

- (a) $y = Ce^{\int P dx}$
- (b) $y = Ce^{-\int P dx}$
- (c) $x = Ce^{-\int P dy}$
- (d) $x = Ce^{\int P dy}$

Solution:

- (b) $y = Ce^{-\int P dx}$

Hint:

Solution of $\frac{dy}{dx} + p(x)y = 0$, Here $Q(x) = 0$

$$y e^{\int P dx} = \int Q e^{\int P dx} dx + c y e^{\int P dx} = 0 + c y = c e^{-\int P dx}$$

Question 9.

The integrating factor of the differential equation $\frac{dy}{dx} + y = 1 + yx$ is

- (a) $x e^x$
- (b) e^{xx}
- (c) λe^x
- (d) e^x

Solution:

- (b) e^{xx}

Hint:

$$\frac{dy}{dx} + y = 1 + yx \Rightarrow \frac{dy}{dx} + y(1 - x) = 1 \Rightarrow \frac{dy}{dx} + y(1 - x) = 1 \Rightarrow \int P dx = \int (1 - x) dx = x - \log x \text{ I.F.} = e^{x - \log x}$$

$$\int P dx = e^x - \log x = e^x e^{-\log x} = e^x x$$

Question 10.

The integrating factor of the differential equation $\frac{dy}{dx} + P(x)y = Q(x)$ is x , then $P(x)$

- (a) x
- (b) x^2
- (c) $\frac{1}{x}$
- (d) $\frac{1}{x^2}$

Solution:

- (c) $\frac{1}{x}$

Hint:

$$\frac{dy}{dx} + P(x)y = Q(x) \text{ Given, I.F.} = e^{\int P dx} = x \Rightarrow \log e^{\int P dx} = \log x \Rightarrow \int P dx = \log x$$

$$\text{we know } \int \frac{1}{x} dx = \log x$$

$$\therefore P = \frac{1}{x}$$

Question 11.

The degree of the differential equation (a) 2

$$y(x) = 1 + dydx + 11.2(dydx)^2 + 11.2.3(dydx)^3 + \dots$$

is

(b) 3

(c) 1

(d) 4

Solution:

(c) 1

Hint:

Degree = 1

Question 12.

If p and q are the order and degree of the differential equation

$$\text{when } ydydx + x^3(d^2ydx^2) + xy = \cos x$$

(a) p (b) p=q (c) p>q (d) \mathrm{p} exists and \mathrm{q} does not exist

Solution:

(c) p>q

Hint:

$$ydydx + x^3(d^2ydx^2) + xy = \cos x$$

Order p=2, degree q=1

∴ p>q as 2>1

Question 13.

The solution of the differential equation $dydx + 1\sqrt{1-x^2} = 0$

(a) $y + \sin^{-1}x = c$

(b) $x + \sin^{-1}y = 0$

(c) $y^2 + 2\sin^{-1}x = c$

(d) $x^2 + 2\sin^{-1}y = 0$

Solution:

(a) $y + \sin^{-1}x = c$

Hint:

$$dydx + 1\sqrt{1-x^2} = 0 \Rightarrow dydx = -1\sqrt{1-x^2} \Rightarrow dy = -\int dx\sqrt{1-x^2} \Rightarrow y + \sin^{-1}x = c$$

Question 14.

The solution of the differential equation $dydx = 2xy$ is

(a) $y = ce^{x^2}$

(b) $y = 2x^2 + c$

(c) $y = ce^{-x^2} + c$

(d) $y = x^2 + c$

Solution:

(a) $y = ce^{x^2}$

Hint:

$$dydx = 2xy \Rightarrow \int \frac{dy}{y} = 2 \int x dx \Rightarrow \log y = x^2 + c \Rightarrow y = e^{x^2+c} = e^{x^2} \cdot e^c \Rightarrow y = ce^{x^2}$$

Question 15.

The general solution of the differential equation $\log(dydx) = x + y$

(a) $e^x + e^y = c$

(b) $e^x + e^{-y} = c$

(c) $e^{-x} + e^y = c$

(d) $e^{-x} + e^{-y} = c$

Solution:

(b) $e^x + e^{-y} = C$

Hint:

$$\log(dydx) = x + y \Rightarrow dydx = e^{x+y} = e^x \cdot e^y \Rightarrow \int e^{-y} dy = \int e^x dx - e^{-y} = e^x + C \Rightarrow e^x + e^{-y} = C$$

Question 16.

The solution of $dydx = 2y - x$ is

Solution:

(c) $12x - 12y = C$

Hint:

$$dydx = 2y - x \Rightarrow dydx = 2y \cdot 2x \int 2 - y dy = \int 2 - x dx - 2 - y \log 2 = -2 - x \log 2 + C$$

Simplifying $12x - 12y = C \log 2 \Rightarrow 12x - 12y = C$ (constant)

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Question 17.

$$dydx = yx + \phi(yx)\phi(yx)$$

The solution of the differential equation is

(a) $x\phi(yx) = k$

(b) $\phi(yx) = kx$

(c) $y\phi(yx) = k$

(d) $\phi(yx) = ky$

Solution:

(b) $\phi(yx) = kx$

Hint:

$dydx = yx + \phi(yx)\phi(yx)$ is a Homogeneous differential equation.

$$dydx - yx + \phi(yx) = \phi(yx)\phi(yx) = kx \quad [\because k \text{ constant}]$$

Question 18.

If $\sin x$ is the integrating factor of the linear differential equation $dydx + Py = Q$, then P is

(a) $\log \sin x$

(b) $\cos x$

(c) $\tan x$

(d) $\cot x$

Solution:

(d) $\cot x$

Hint:

$$dydx + Py = Q$$

$$\text{I.F.} = e^{\int P dx} = \sin x \Rightarrow \log e^{\int P dx} = \log \sin x \Rightarrow \int P dx = \log \sin x$$

$$\text{we know } \int \cot x dx = \log \sin x \Rightarrow \therefore P = \cot x$$

Question 19.

The number of arbitrary constants in the general solutions of order n and $n+1$ are respectively

(a) $n-1, n$

(b) $n, n+1$

(c) $n+1, n+2$

(d) $n+1, n$

Solution:

(b) $n, n+1$

Question 20.

The number of arbitrary constants in the particular solution of a differential equation of third order is

- (a) 3
- (b) 2
- (c) 1
- (d) 0

Solution:

- (d) 0

Question 21

Integrating factor of the differential equation

$$dy/dx = x + y + 1$$

- (a) $1x+1$
- (b) $x+1$
- (c) $1\sqrt{x+1}$
- (d) $\sqrt{x+1}$

Solution:

- (a) $1x+1$

Hint:

$$dy/dx = x + y + 1 \Rightarrow dy/dx - y = x + 1 \Rightarrow dy/dx - y = 1(x+1) \Rightarrow \int P dx = -\int 1(x+1) dx = -\log(x+1) = \log(1/x+1) \text{ I.F.} = e^{\int P dx} = e^{\log(1/x+1)} = 1/x+1$$

Question 22.

The population P in any year t is such that the rate of increase in the population is proportional to the population. Then

- (a) $P = ce^{kt}$
- (b) $P = ce^{-kt}$
- (c) $P = ckt$
- (d) $P = c$

Solution:

- (a) $P = ce^{kt}$

Hint:

$$\text{Given } dp/dt \propto P \Rightarrow dp/dt = kP \Rightarrow \int dp/P = \int k dt \Rightarrow \log P = kt + c \Rightarrow P = e^{kt+c} = ce^{kt}$$

Question 23.

P is the amount of certain substance left in after time t . If the rate of evaporation of the substance is proportional to the amount remaining, then

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- (a) $P = ce^{kt}$
- (b) $P = ce^{-kt}$
- (c) $P = ckt$
- (d) $Pt = c$

Solution:

- (b) $P = ce^{-kt}$

Hint:

$$\text{Given } dp/dt \propto P$$

$$dp/dt = -kP \Rightarrow \int dp/P = \int -k dt \Rightarrow \log P = -kt + c \Rightarrow P = e^{-kt+c} = ce^{-kt}$$

[decay constant]

Question 24.

If the solution of the differential equation $dy/dx = ax + 32y + f$ represents a circle, then the value of a is.....

- (a) 2
- (b) -2

(c) 1

(d) -1

Solution:

(b) -2

Hint:

$$dy/dx = ax + 3 \Rightarrow \int dy = \int (ax + 3) dx \Rightarrow y = \frac{ax^2}{2} + 3x + c \Rightarrow ax^2 - 2y + 3x - 2c = 0$$

In the equation of circle,

Co efficient of ' x^2 ' = Co efficient of ' y^2 '

$$a = -2$$

Question 25.

The slope at any point of a curve $y=f(x)$ is given by $dy/dx=3x^2$ and it passes through $(-1,1)$.

Then the equation of the curve is

(a) $y=x^3+2$

(b) $y=3x^2+4$

(c) $y=3x^3+4$

(d) $y=x^3+5$

Solution:

(a) $y=x^3+2$

Hint:

$$dy/dx = 3x^2 \Rightarrow \int dy = \int 3x^2 dx \Rightarrow y = x^3 + c$$

It passes through $(-1,1)$

$$\therefore 1 = -1 + c \Rightarrow c = 2$$

$\therefore$ Equation of the curve is $y=x^3+2$